

Classification & Logistic Regression II

Prepared by

Dr. Yasmeen Khedr

**Assistant Professor of Information Technology Dept., Faculty of Computer
& Informatics , Zagazig University**

PhD, Huazhong University of Science and Technology, Wuhan, China



Outline

1

Classification

2

Linear Classifiers

3

Logistic Regression

4

Sigmoid Function

5

Loss Function

6

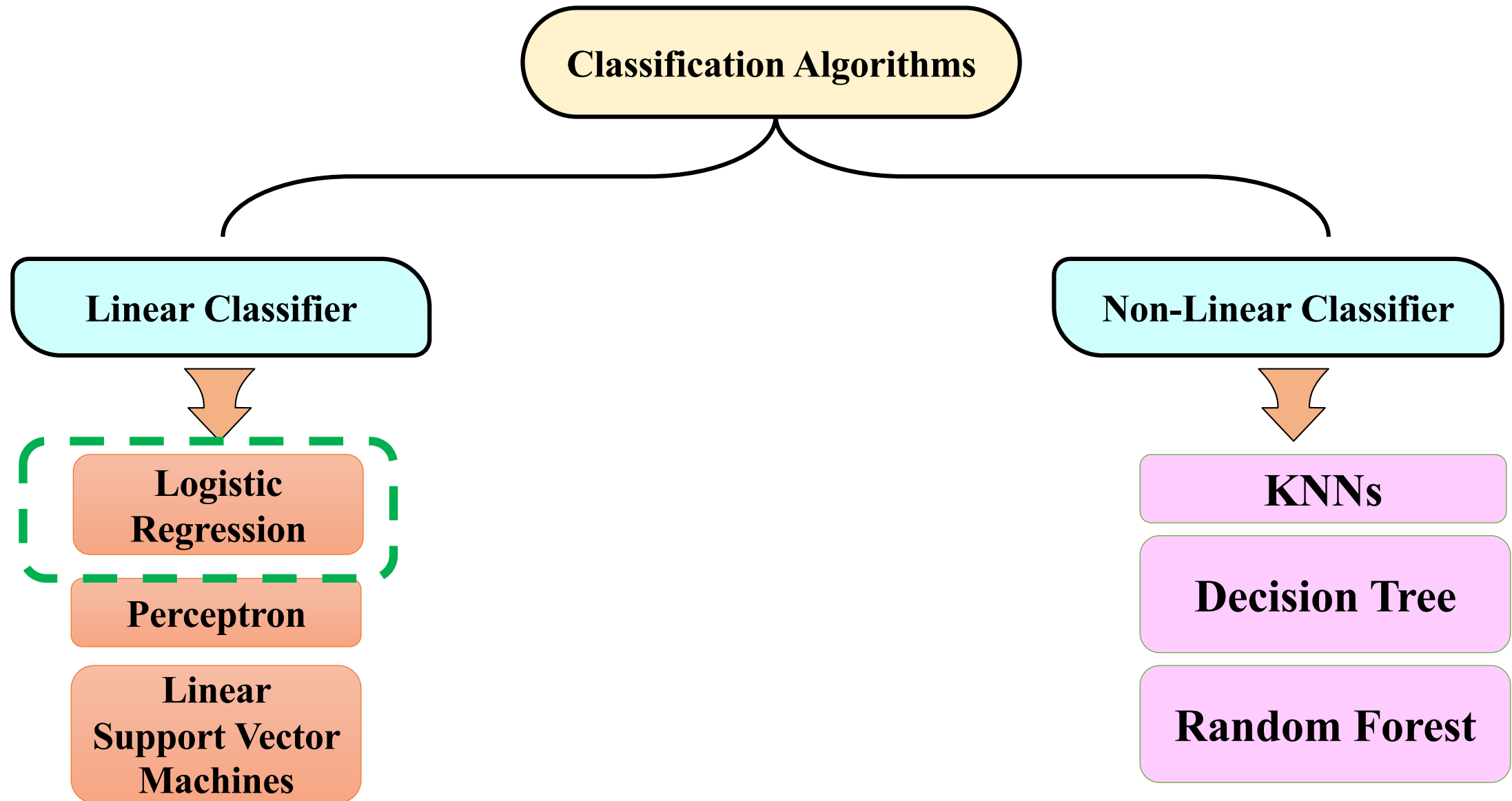
Gradient Descent for Classification

7

Softmax Regression

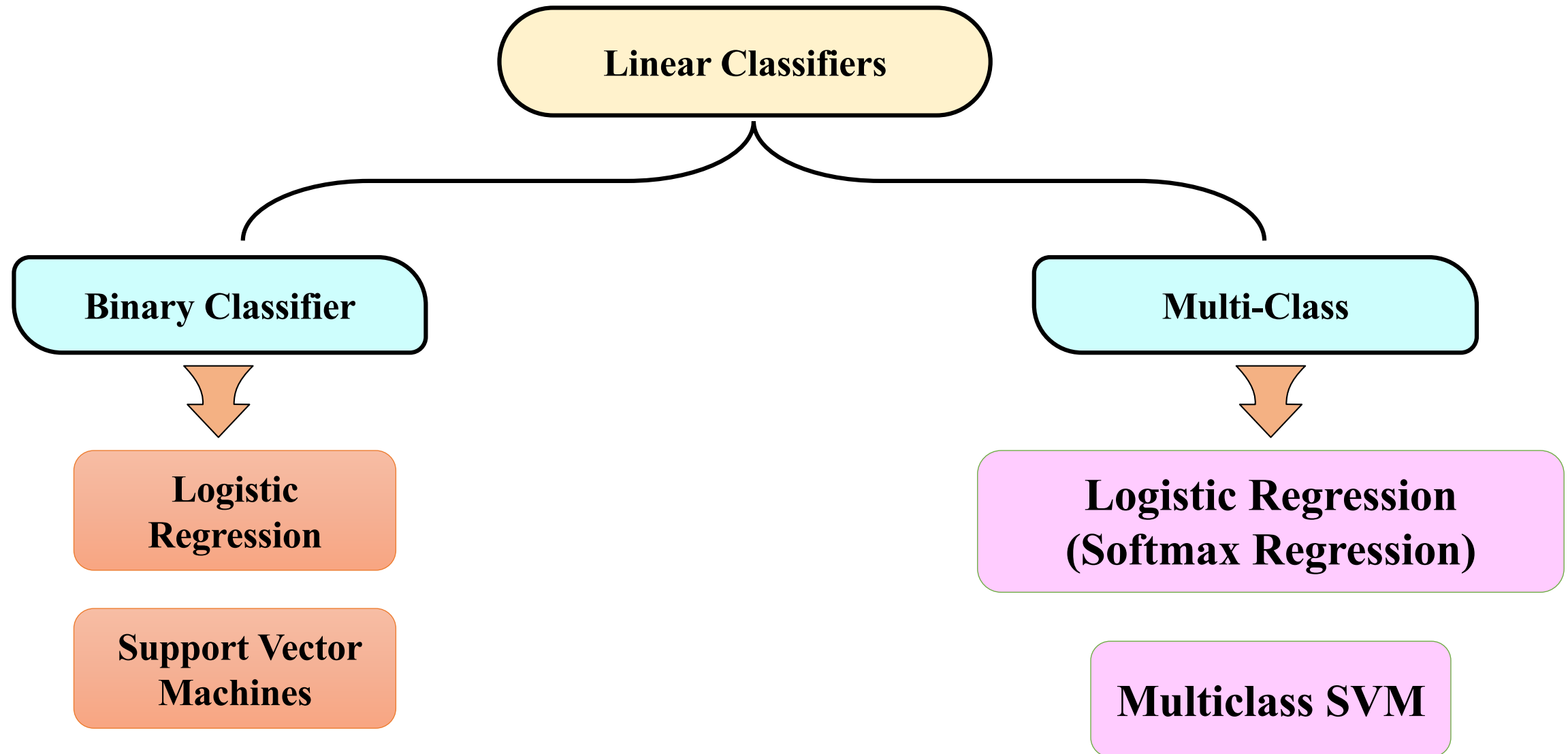
1. Classification

Type of Algorithms



2. Linear Classifier

Type of Algorithms



2. Linear Classifier

Overview

Goal: Find a linear decision boundary to separate classes.

Class prediction: Binary $\text{sign}(f(x))$, Multi-class $\text{argmax } f(x)$

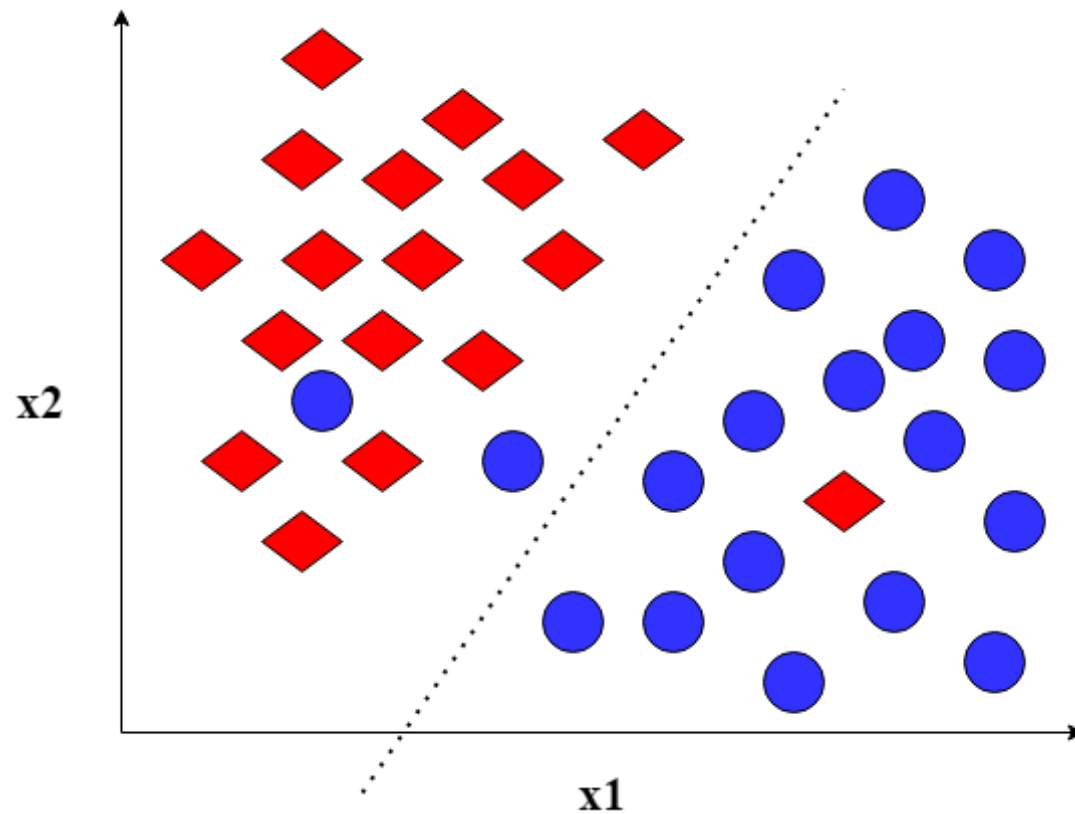
Linear: model prediction y is a linear function of x , followed by a threshold r :

$$z = \mathbf{w}^\top \mathbf{x} + b$$
$$y = \begin{cases} 1 & \text{if } z \geq r \\ 0 & \text{if } z < r \end{cases}$$

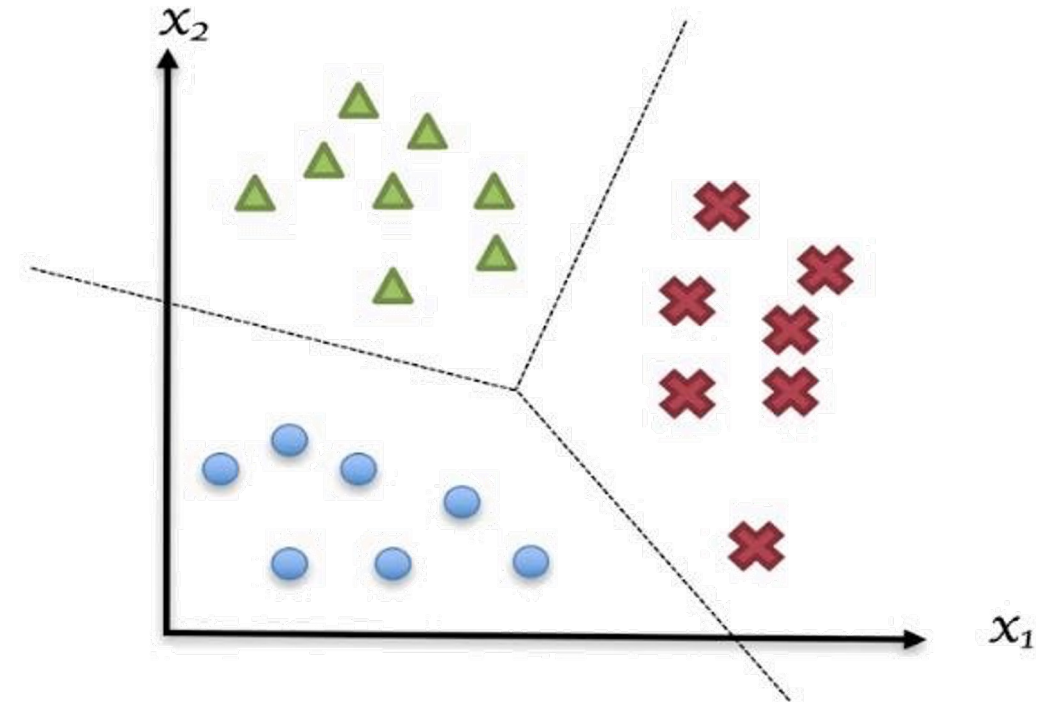
2. Linear Classifier

Overview

Binary Linear Classifier



Multiclass Classifier

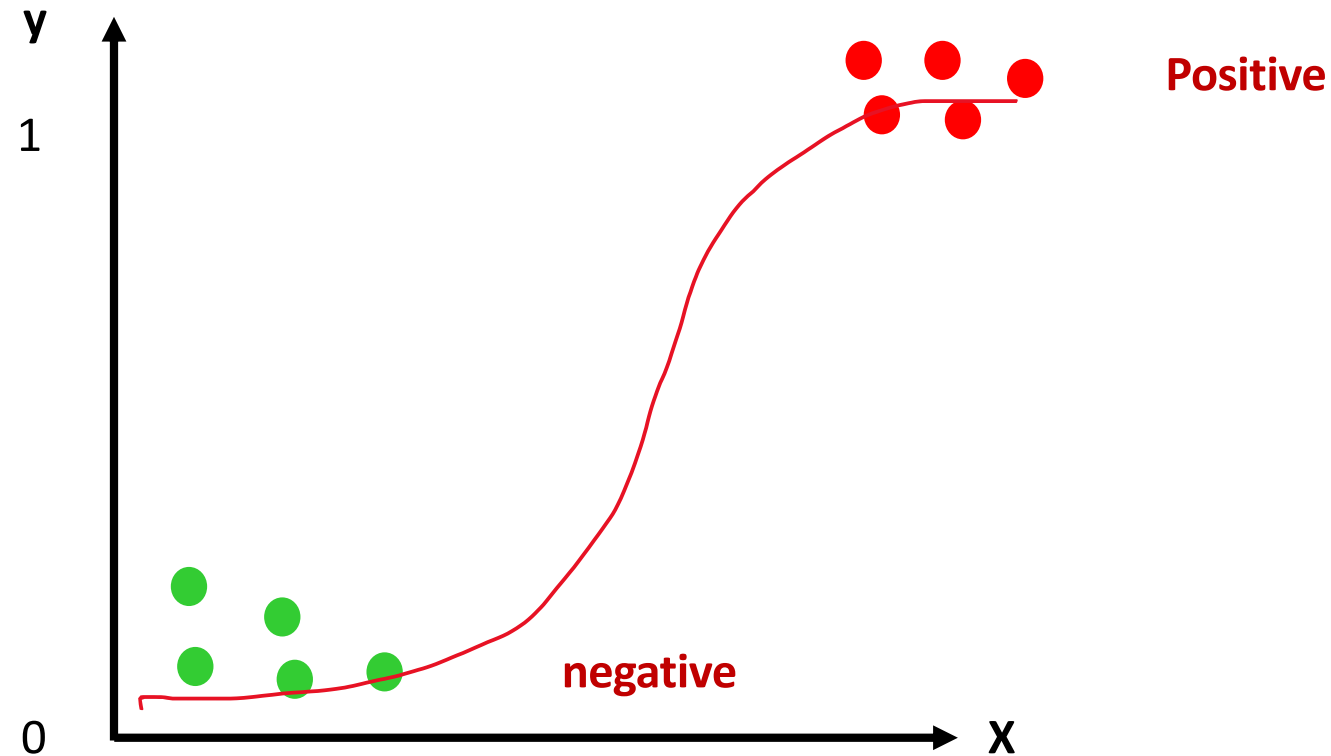


3. Logistic Regression

Logistic Regression

Output between 0 and 1

To solve this problem, we use Sigmoid function



4. Sigmoid Function

Sigmoid Function

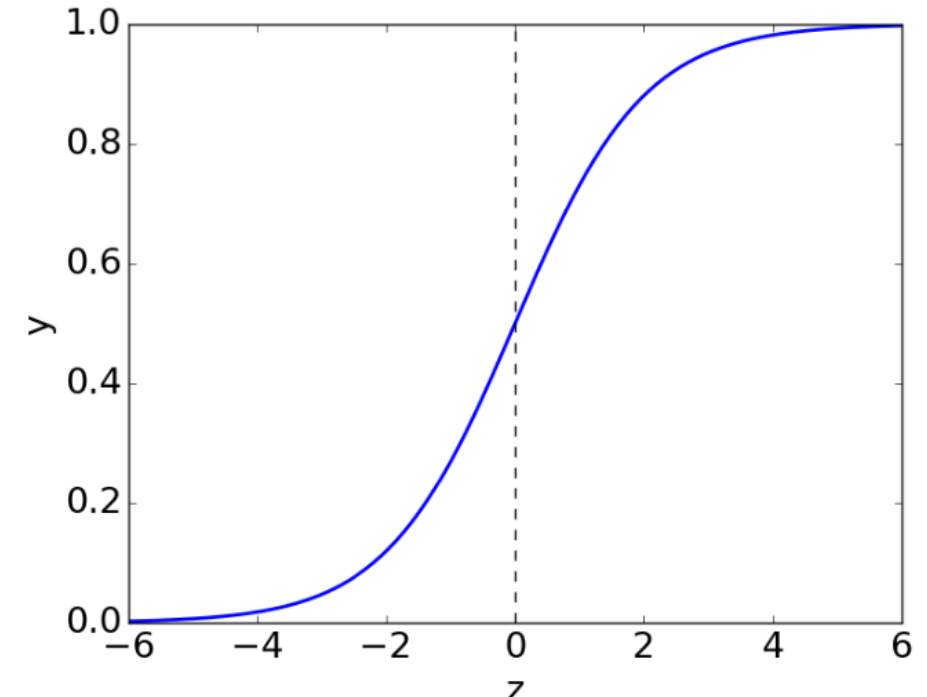
Sigmoid function is called S-shaped function (activation function)

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

where e is the base of the natural logarithm

$$z = \mathbf{w}^T \mathbf{x} + b$$

$$h = \sigma(z)$$



4. Sigmoid Function

Sigmoid Function

When Z is negative:

- The exponent (e^{-Z}) becomes **a large positive value**.
- Thus, the denominator increases, making σ (the sigmoid output) small (close to 0).

When Z is very small negative:

- The exponent (e^{-Z}) becomes an **extremely large positive number**.
- The denominator dominates, so $\sigma \approx 0$.

4. Sigmoid Function

Sigmoid Function

When Z is positive:

- The exponent (e^Z) becomes **a small negative value**.
- Thus, the denominator decreases, making σ increase toward 1.

When Z is very large positive:

- The exponent (e^Z) becomes an **extremely large negative number**.
- The denominator dominates, so $\sigma \approx 1$.

4. Sigmoid Function

Sigmoid Function

Example:

x is temperature

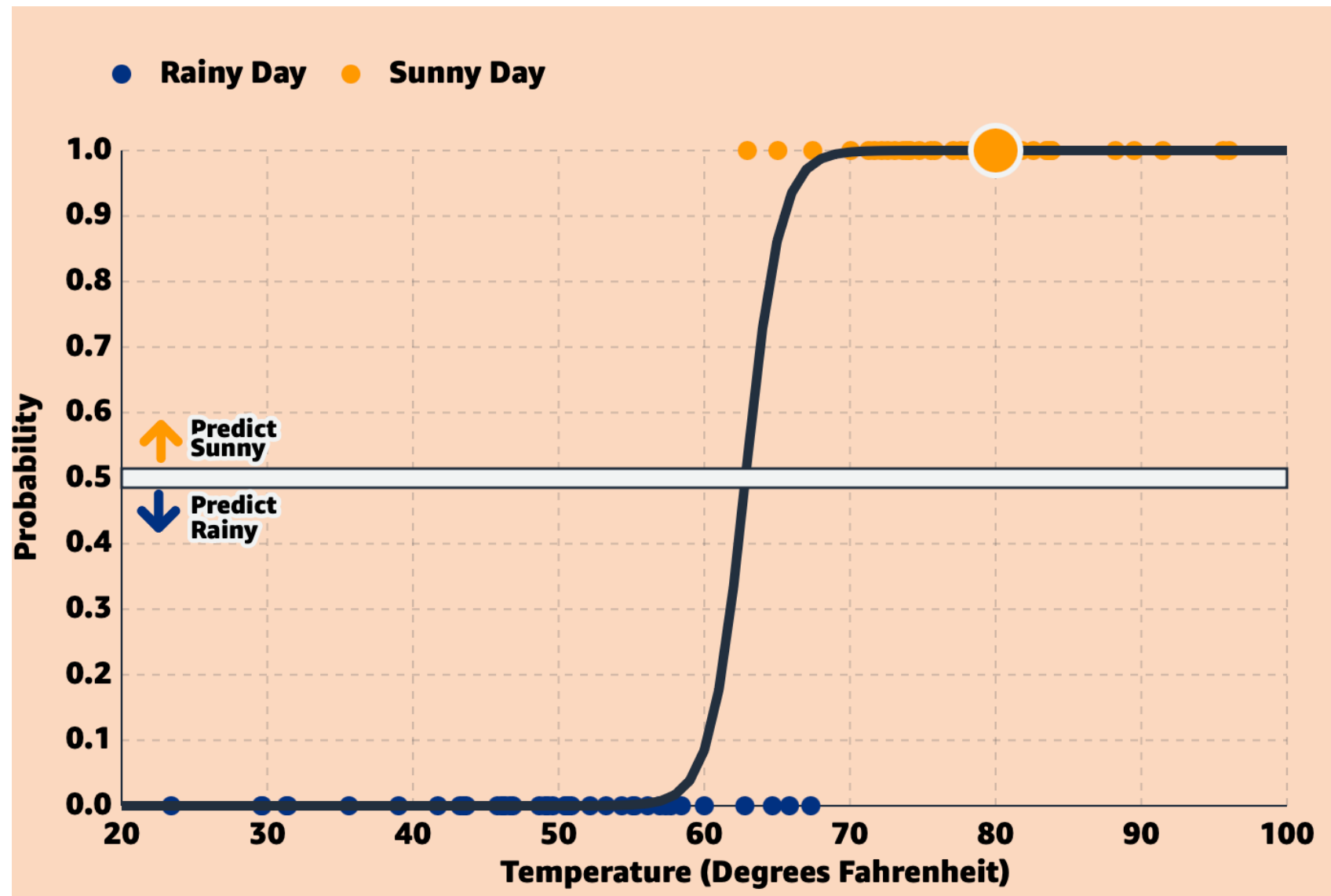
y is 0 to a Rainy Day or
1 to a Sunny Day

$R = 0.5$

$X = 80$

$Y = 0.9 = 1$

90% chances y is 1



5. Loss Function

Cross-entropy loss

$$J(w, b) = \frac{1}{n} \sum_{i=1}^n \text{cost}(h_i - y_i)$$

$$\text{cost}(h_i - y_i) = -\log(h_i) \quad \text{if } y = 1$$

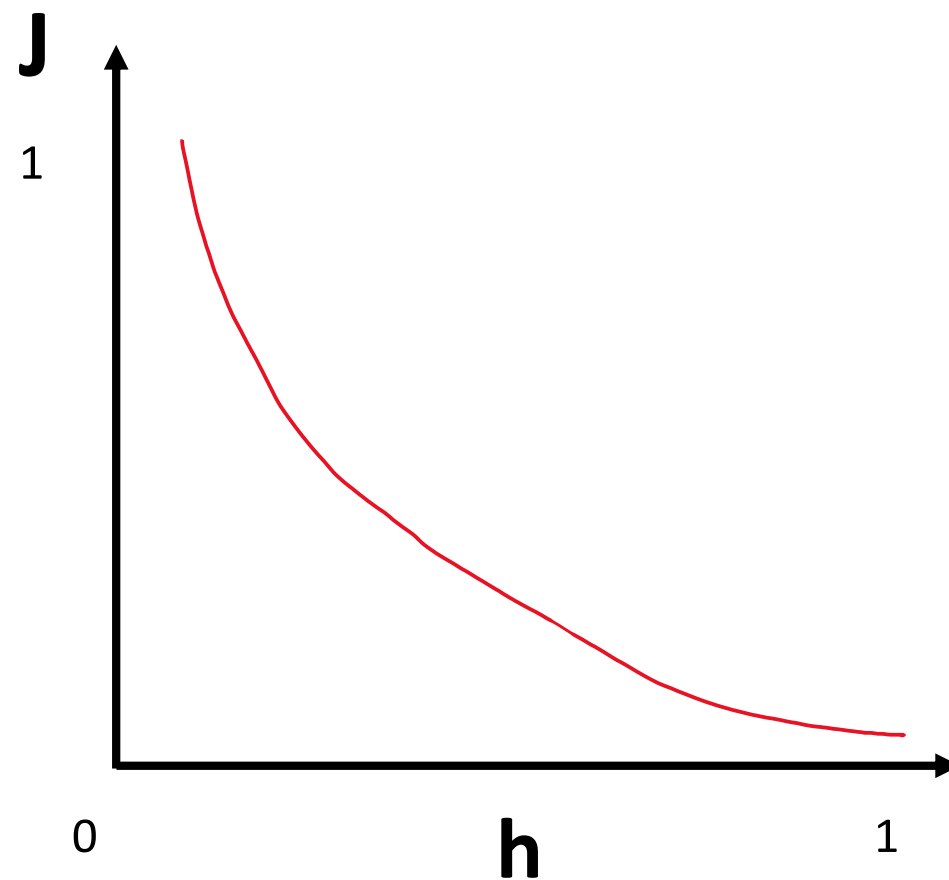
$$\text{cost}(h_i - y_i) = -\log(1 - h_i) \quad \text{if } y = 0$$

5. Loss Function

Cross-entropy loss

$$y = 1$$

$$\text{cost}(h_i - y_i) = -\log(h_i)$$

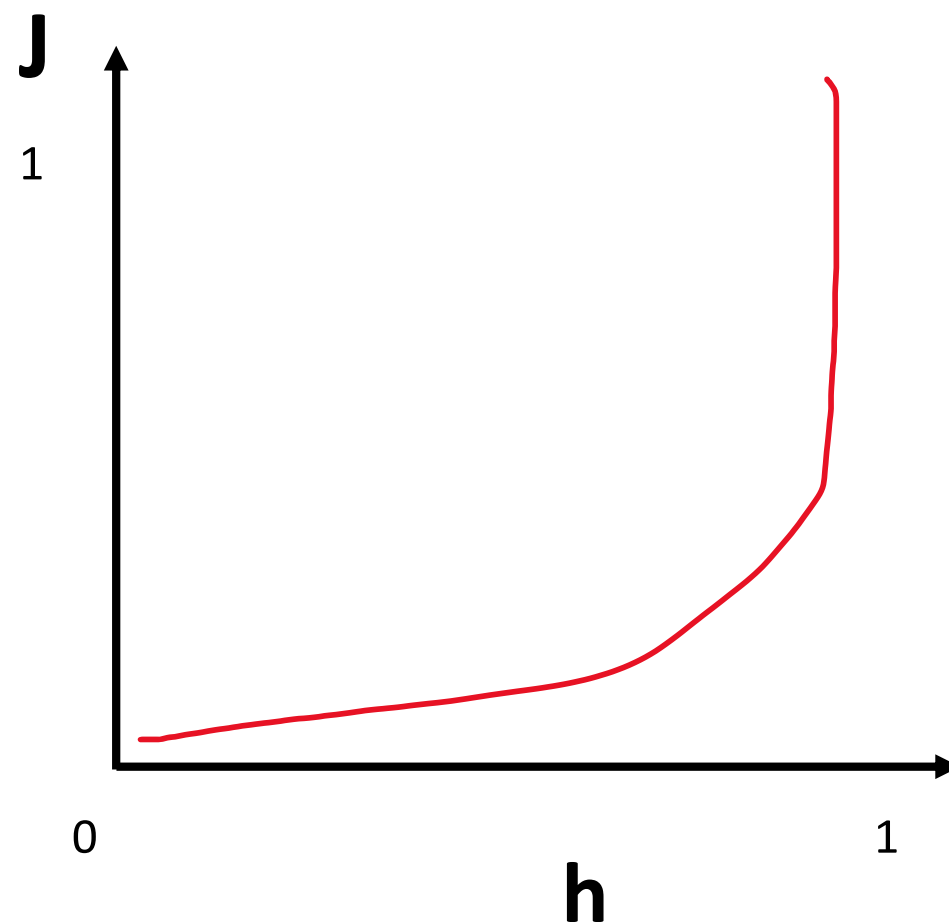


5. Loss Function

Cross-entropy loss

$$y = 0$$

$$\text{cost}(h_i - y_i) = -\log(1 - h_i)$$



5. Loss Function

Cross-entropy loss

$$J(\mathbf{w}, b) = -\frac{1}{N} \sum_{i=1}^N [y_i \log(h_i) + (1 - y_i) \log(1 - h_i)]$$

Where:

$$h_i = \sigma(\mathbf{w}^T \mathbf{x}_i + b) = \frac{1}{1 + e^{-(\mathbf{w}^T \mathbf{x}_i + b)}}$$

y_i is actual value (true value) $\in [0,1]$

N is number of samples

5. Loss Function

Cross-entropy loss

$$J(\mathbf{w}, b) = -\frac{1}{N} \sum_{i=1}^N [y_i \log(h_i) + (1 - y_i) \log(1 - h_i)]$$

Chain Rule: Split $\frac{\partial J}{\partial \mathbf{w}}$ into 3 sub-derivatives.

Term 1: Compute $\frac{\partial J_i}{\partial h_i}$ (log loss derivative).

Term 2: Compute $\frac{\partial h_i}{\partial z_i}$ (sigmoid derivative).

Term 3: Compute $\frac{\partial z_i}{\partial \mathbf{w}}$ (linear term derivative).

Combine: Cancel terms to get $(h_i - y_i)\mathbf{x}_i$.

6. Gradient Descent Algorithm

Gradient Derivation

We compute the gradient of the cost function

$$\frac{\partial J}{\partial \mathbf{w}} = \frac{1}{N} \sum_{i=1}^N (h_i - y_i) \mathbf{x}_i, \quad \frac{\partial J}{\partial b} = \frac{1}{N} \sum_{i=1}^N (h_i - y_i)$$

Update rule:

$$\mathbf{w} \leftarrow \mathbf{w} - l \cdot \frac{\partial J}{\partial \mathbf{w}}$$

$$b \leftarrow b - l \cdot \frac{\partial J}{\partial b}$$

7. Softmax Regression

Cross-entropy loss for Multi-Class

$$J(\mathbf{W}, \mathbf{b}) = -\frac{1}{N} \sum_{i=1}^N \sum_{k=1}^K y_{i,k} \log(p_{i,k})$$

Where:

$$p_{i,k} = \frac{e^{z_{i,k}}}{\sum_{j=1}^K e^{z_{i,j}}}, \text{ and } z_{i,k} = \mathbf{w}_k^T \mathbf{x}_i + b_k.$$

y_i is actual value (true value)

N, k is number of samples

7. Softmax Regression

Gradient Derivation

We compute the gradient of the cost function

$$\frac{\partial J}{\partial \mathbf{w}_k} = \frac{1}{N} \sum_{i=1}^N (p_{i,k} - y_{i,k}) \mathbf{x}_i$$

Update rule:

$$\mathbf{w} \leftarrow \mathbf{w} - l \cdot \frac{\partial J}{\partial \mathbf{w}}$$

$$b \leftarrow b - l \cdot \frac{\partial J}{\partial b}$$

A thick blue rectangular frame surrounds the central text. At each of the four corners, there is a small cluster of three blue squares of varying sizes, arranged in a stepped pattern.

Thank you